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A Super-Harvest System for NSA?

by

W. R. Elmendorf

This is a cursory analysis of the need for a Super-Harvest system by the National Security Agency for its information processing activities. No contact was made with NSA; the limited information to be presented was garnered from conversations with IBM people familiar with the problems of this agency. None of the material presented herein was obtained from classified sources.

Nature of the Problem

While "the" problem is naturally a collection of many different problems, the great bulk of these have a common flavor. This flavor is complex combinatorial analysis of vast quantities of information. This implies re-ordering, correlation, and frequency distribution. File maintenance and editing are also important. There is great repetitiveness in the processing. The classes of processing operations are expected to remain relatively unchanged over the years.

The nature of the problems was well stated in this excerpt from a report[2] on Harvest:

> ". . . . . the information is either non-numeric or is developed within areas supported by little or no theoretical structure . . . . . These areas are in the early stages of scientific development where ordering and classification are predominant activities and where the major problem is to uncover patterns and trends upon which theory can be built. There are few rules for determining the relevancy of information, so that often enormous amounts must be examined to achieve

> significant results. Comparatively simple operations and data transformations are the most useful, and the output desired may frequently be some statistical characterization of the input."

There is no nearby upper bound on this agency's need for information processing capacity.

Programming

This is an especially significant problem in the application of such a large, complex, and novel system. In a typical week about 500 separate batch type, off-line jobs are run. About a third of these are periodic. Another third are routine sorting runs, program assembly, and the like, which use general utility programs. The remaining third are one-time, specially programmed jobs. (These percentages relate to jobs - not to running time).

There is an undercurrent of disagreement over the programming philosophy of the Harvest Streaming Unit. One school favors less generality in the logical capabilities of this unit; the other school favors greater generality. In any case, it seems clear that full utilization of the Streaming Unit is a task for experienced and imaginative programmers; such people are not easy to find. This dilemma is not uncommon and must be reckoned with in the design of future systems.

Machine Organization Conjecture

On-line operation - that is, direct connection of the processor to communication lines - would possibly prove valuable in special situations demanding fast response.

There is a difference of opinion as to the merit of an intimate man-machine communication link. Such a linkage would permit in-line program branching conditional on human decisions made during the course of program execution. This idea is questioned by some because the analyst's decisions involve wholesale discard of processing already done, as well as giving direction to future processing, and because the points at which human intervention would be worthwhile are infrequent relative to the processing time needed to generate the necessary decision information. Nevertheless this mode of operation seems worthy of further investigation.

An associative memory - even a small one - appears to have particular appeal. The NSA problems involve scanning of high volumes of information

to discern patterns in the appearance of hypothesized arguments. Consequently, it would be helpful to be able to compare in one step a given argument against a quantity of information, to record the associations ("less than", "greater than", "between", as well as "equal to") in a "hit" register, and then to analyze the pattern of hits which results. (In Harvest considerable streaming and sequential logical testing would be necessary to achieve the indicative information developed in one step in the hit register.) Following this action a new argument could be substituted and the process repeated without changing the information in the associative memory. Thus, it is possible to stream the arguments without streaming the data, and to store a new batch of data in the associative memory only when the stream of arguments has expired.

A more sophisticated approach would involve the use of multiple associative memories (conceptually, if not physically) - all containing the same data but with the argument registers containing different arguments. Now it is possible to cross-analyze the patterns developed in the different hit registers to discern second level patterns. Following this action the arguments could all shift to the argument register of the succeeding memory module, thus dropping one argument and introducing a new one. Broadside association of the slightly altered set of arguments would then introduce changes in the hit pattern, which would be analyzed anew. Cycling in this way suggests a streaming of arguments past multiple unchanged associative memories. This concept is illustrated in Figure 1. Sixteen 64 word associative memories would be useful in this type of processing.

A natural extension of Harvest design would allow for connection of multiple streaming units - some specialized and some general purpose - with concurrent operation of all units. If the units were logically cascaded then vertical processing simultaneity could be achieved. This would appear advantageous only where an imbalance exists between the memory/bus system flow rate and the streaming unit flow rate.

A general approach to the combinatorial analysis problem would seem to be use of a technique for construction of strings of data elements through inductive reasoning based on various pieces of element ordering evidence. Such a process is described by H. Fleisher in a separate report.

Any large data processing installation (and this customer reportedly has one of the largest) is composed of an assemblage of discrete pieces of equipment collected over a period of time from different manufacturers. Some will be quite general purpose and others quite special purpose. Some will be controlled by stored programs, some by wired programs, and some by built-in programs. However, neither

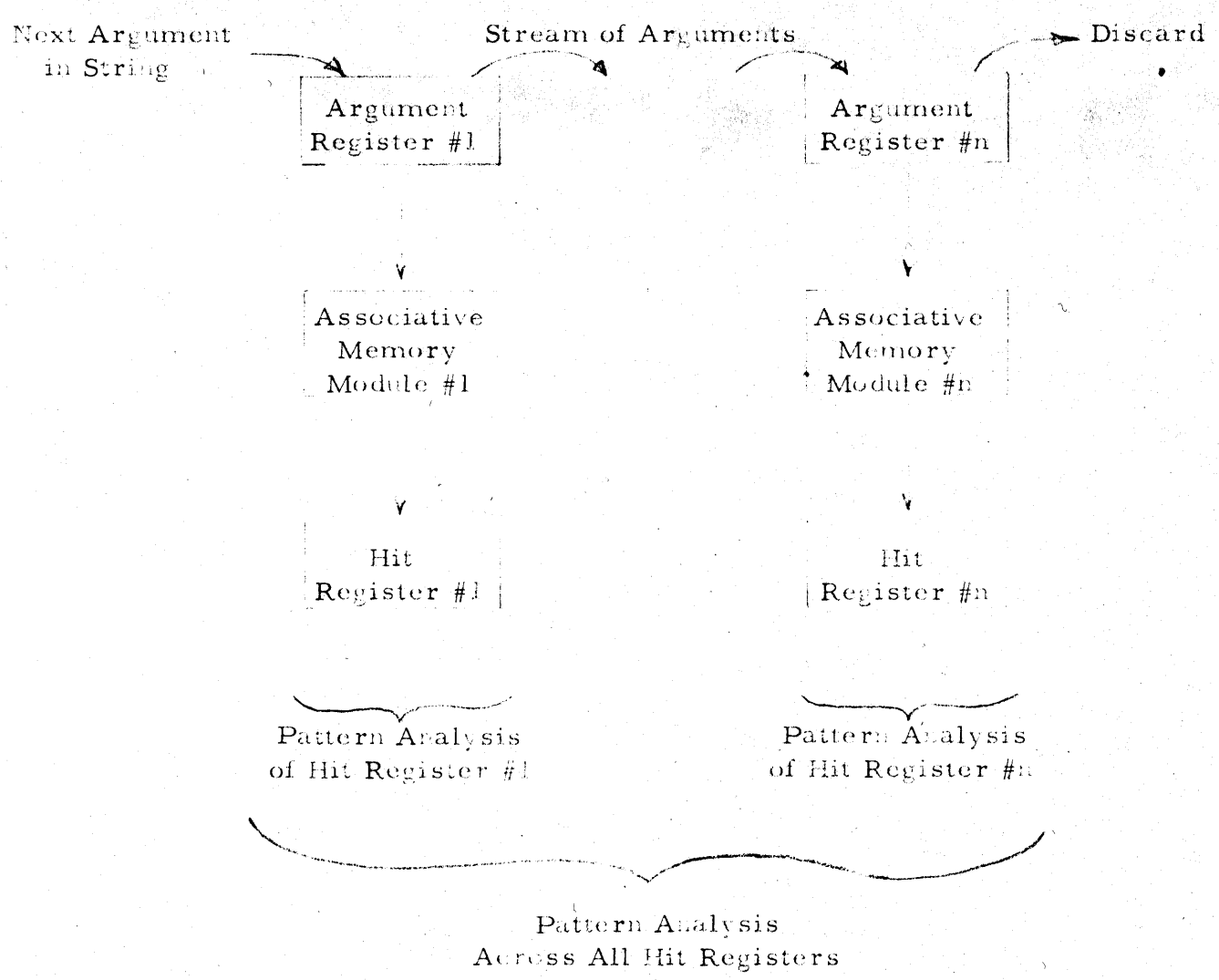
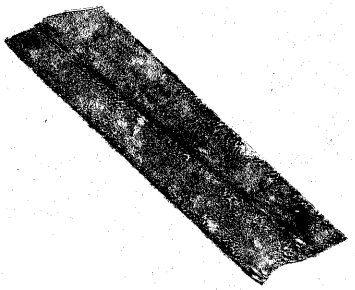


Figure 1 - Suggested Use of Multiple Associative Memories in Pattern Analysis



the overall programs nor the data are inherently broken up into discrete chunks which apply uniquely to individual units of the machine complex. Consequently, program control and data must pass from unit to unit during the course of processing. Such a machine complex has two characteristics: data does not flow between units electronically, and program control does not flow between units electronically. The integration of data and control flow is effected by the people who operate the system. It is conjectured that electronic integration of the machine complex (multi-processing) would be a worthwhile improvement.

Recommendations

It is apparent that NSA is, and will continue to be, a large information processing organization with the need, interest, and money to support advanced machine projects. Consequently, if the professional ties between NSA and IBM are amenable to a cooperative effort, a continuing awareness of, and response to, the agency's problems would certainly prove beneficial to both parties and would advance the information processing art.

Solidly rooted ideas for an improved information processing system are not possible without a clear understanding of the work to be done by the system. This means a good knowledge of the current system, plus the vision to foresee future trends. Certainly the former, and probably the latter, are dependent on one's freedom to gather information and engage in frank discussion with knowledgeable people. In this instance a high level of security clearance is a prerequisite to such discussion. This study was made without benefit of such clearance; any future study-in-depth must have it.

W. R. Elmendorf

References

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4. David Kahn, "Lgcn Otuu Wllwgh Wl Etfown," New York Times Magazine, November 13, 1960, beginning on p. 71.